

various warfare revolutions in the history of mankind.

Warfare revolutions

Dogs and horses were probably the first animals used in war. Many animals are still being used by modern military and police forces. We learn from history that animals, mammals, birds, etc have also been reportedly used in wars in the earlier days. But modern-day warfare revolutions are largely driven by technology, and the ability to achieve precision and efficiency. The word revolution itself may be controversial depending on how you perceive it. However, each revolution has an element of uniqueness and disruptive nature.

The first revolution goes back to the invention of gun powder and the use of automatic machine guns in the first world war (WWI). Battle tanks were developed, and these carried fire guns across the battlefield along with foot soldiers on the ground. War planes were also developed during this period. The battle was fought between military units.

The second revolution saw the invention of the nuclear bomb, which was used in the second world war (WWII). This revolution witnessed much more powerful warfare that caused mass destruction because no particular military unit was targeted. Fire power had the power to wipe off the entire targeted area or even an entire city. Jet engines, ballistic missiles and submarine nuclear warheads were also developed during the second revolution.

The third revolution includes the use of drones, unmanned aerial vehicles (UAVs) and AI. With the use of AI technology, earlier electronics warfare techniques are becoming more intelligent, effective and precise in many ways.

Advances in AI technology have led to lethal autonomous weapons (LAWs), where robots can independently search and engage targets without human intervention. Many AI-based LAWs have been developed worldwide. Experts predict that with advancements in technology, AI and robots could eventually replace soldiers in warfare. Thus,



BAE Systems Taranis drone (Credit: Wikipedia)

many people think that AI is the third revolution in warfare.

Again, in recent decades we have witnessed the ever-prevalent use of computers and AI technology in warfare. With the emergence of AI, there is a threat of cyber warfare and autonomous weapons. The very facts that we are living in a connected world now and that Internet-based equipment are used by defence departments make us vulnerable to cyber-attacks.

Militaries around the world are upgrading their nuclear weapons with greater automation and connectivity. Digital automation and connectivity are essential in today's defence strategies and operations. For example, a submarine nuclear warhead deployed in the ocean depends on different computer systems, such as the nuclear propulsion plant, navigation, fresh air, fresh water and so on. The different computer systems, including codes, are written and built by human experts. This could mean that an enemy can hack the systems.

Cyber-attacks can also paralyse major critical national infrastructures, such as power distribution grids, transportation, information hubs, etc. Hence, all cyber-attacks require quick and effective decision-making. In case of delayed decisions, there can be chaos and confusion about social order, which can make it difficult to initiate damage control and restore normalcy. It could also disrupt traditional systems and result in more predictable patterns of conflict escalation, leading to war including nuclear attacks as the worst-case scenario.

Why AI could be dangerous

As per Wikipedia, the deadliest war in history, in terms of the cumulative

number of deaths since its start, is WWII, with 60 to 85 million deaths, followed by the Mongol conquests at up to 60 million deaths. Now, as technology progresses, the world is marching towards more dangerous situations. While it is true that AI technology can radically help improve the precision, effectiveness and efficiency of many weapons,

development of LAWs could be dangerous.

AI-based LAWs can be a scary thing in future because of two main reasons. First, AI could go wrong and out of control. In 2007, in South Africa, a semi-autonomous robotic cannon went out of control and killed nine friendly soldiers and injured fifteen others on the spot.

Second, an AI system could be hacked by a rogue state or terrorist, trigger deadly weapons and cause devastation to entire cities.

Use of AI technology for military applications including defensive, offensive and information warfare is increasing day by day. On defensive systems, LAWs can autonomously identify and attack oncoming missiles, rockets, artillery fire, aircraft and surface vessels according to the criteria set by the human operator. Many missile defence systems, such as Iron Dome, also have autonomous targeting capabilities. Examples of offensive systems include stationary sentry guns, which can fire at humans and vehicles, being used in South Korea and Israel.

With the rolling out of 5G networks, there will be more bandwidth and higher speed of digital operations and automation, further enabling the rapid rise of AI technology. This also raises serious concerns, because apart from self-driving cars and remote robotic surgeries, the world is susceptibility to cyber-attacks and surveillance, in addition to development of military autonomous vehicles and weapons. Therefore without proper regulation, the world is moving towards a dangerous situation in this era of AI technology.